



ANALYSIS REPORT

10398871.r1.v2 NUMBER

Malware Analysis Report

2022-10-13 DATE

Notification

- Author : github.com/miho030
- TLP Classification : TLP:CLEAR
- Purpose : Personal development and public information sharing (non-commercial)

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Summary

Description

The Windows executable contained in the malicious image file being analyzed is, BotNet client program named '[njRAT](#)' (*Bladabindi/NjwOrm*). It has the most educational information and guides among the BotNet tools distributed in the market, and is one of the easily available RAT attack tools. njRAT is known for being an easy-to-reach attack tool for many novices or attackers trying to get info hacking technology because there are many instructions and lecture materials about this BotNet on Youtube or elsewhere.

It is one of the tools favored by attackers since it was first confirmed to have been used by the middle Eastern country on January 1, 2013,

njRAT is configured based on the '[.NET framework](#)'. This RAT tool allows an attacker to remotely control the victim's system, and its main functions include Keyboard Sniffing, Webcam record/streaming, and Steel credentials (Browser/programs) Attacker can

access the system command tool to manipulate the victim system’s processes and remotely upload/download/execute files. It also includes the ability to steal cryptocurrency and account information held by the victim’s cryptocurrency wallet, credit card information registered in the account by targeting cryptocurrency application operating on the victim’s system. additionally, attacker can get victim’s payment information (e.x. credit card number or something else) and victim’s account credentials via browser store data.

Submitted Files (1)

302e1152c2710ad56ef1ff5536bb6539f6ac6901caba28954c2b160f9c7ac518 (Server.exe)

IPs (1)

188.242.138.137

Findings

302e1152c2710ad56ef1ff5536bb6539f6ac6901caba28954c2b160f9c7ac518

Tags

trojan rat njrat bladabindi exe

Details

Verdict	Malware activity
File name	Server.exe
File size	93.00 KB (95232 bytes)
File type	PE32 Win32 EXE
Magic	PE32 executable (GUI) Intel 80386 32-bit Mono/.Net assembly, for MS Windows
MIME	application/vnd.microsoft.portable-executable
MD5	f2d7a6c71e5e63bffa256d2d73eccc7
SHA256	302e1152c2710ad56ef1ff5536bb6539f6ac6901caba28954c2b160f9c7ac518
ssdeep	768:7Y3MxslorpLocxYZJ6aL+2ap0XKC/WpndwruXxrjEtCdnI2piIRz4Rk3BsGdpKgM:9xuo966aL9a+7+d5jEwzGiIdDxOKgS
Entropy	47

Antivirus

Ahnlab	Trojan/Win32.Bladabindi.R295982
Alyac	Generic.MSIL.Bladabindi.F1687BCF
Avast	Win32:KeyloggerX-gen [Trj]
Avira	TR/Dropper.Gen
Bitdefender	Gen:NN.ZemsilF.34738.fiW@a0f5mvj
Fortinet	MSIL/Bladabindi.AS!tr
Kaspersky	HEUR:Trojan.Win32.Generic
MacAfee	BehavesLike.Win32.V80bfus.nm

Malwarebytes	Generic.Worm.Autorun.DOS
symantec	ML.Attribute.HighConfidence

Portable Executable Info

.NET details

- Module Version id	dd2bf338-9409-47bf-94a5-214273745dcd
- TypeLib id	efe9eadc-d4ae-4b9e-b8ab-7e47f8db6ac9

Header

- Target Machine	Intel 386 or later and compatible processors
- Compilation TimeStamp	2021-06-15 21:56:51 UTC
- Entry Point	0x18f3e
- Contained Sections	2

YARA Rules

```
/**
 * njRAT sample - ANY.RUN task 485149f9 (SHA-256: 302E1152...)
 * Created : 2025-05-29
 * Author: github.com/miho030
 * refer docs : "github.com/miho030/Awesome-Malware-Report"
 */
rule njrat_302E1152_generic
{
    meta:
        family           = "njRAT / Bladabindi"
        sha256            = "302E1152C2710AD56EF1FF55368B6539F6AC6901CABA28954C2B160F9C7AC518"
        compiletime       = "2021-06-15 12:56:51"
        reference_1       = "github.com/miho030/Awesome-Malware-Report"
        reference_2       = "ANY.RUN task 485149f9"

    strings:
        $a1 = "Explorer.exe" ascii           /* self copy target file name */
        $a2 = "netsh firewall add allowedprogram" ascii nocase /* MS defender bypass cmd */
        $a3 = "SEE_MASK_NOZONECHECKS" wide ascii           /* signature PATH value */
        $ip = "188.242.138.137" ascii         /* Static C2 IP address */
        $ddns = "aGFraW0z*i5kZG5zM5ldA!!" ascii /* '!!' is for base64 padding */
        $port = "NTU1Mg==" base64

    condition:
        uint16(0) == 0x5A4D /* PE header */
        and pe.timestamp == 0x60C8A393 /* Same TimeStamp with report */
        and filesize < 500KB /* actual sample size range */
        and 3 of ($a*) /* matching with over 3 rules */
        and $ddns /* Include signature C2 domain */
}
```

PE Metadata (EXIF)

Signature	424a5342
-----------	----------

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Created time	2022-08-10 23:23:44
Modified time	2021-06-15 15:56:52
Import Hash	f34d5f2d4577ed6d9ceec516c1f5a744
ImageFile Characteristics	Executable, 32-bit
PEType	PE32
LinkerVersion	8.0
Code size	17000
InitializedDataSize	200
UninitializedDataSize	0
OSVersion	Windows 95 / 4.0 Win NT 4.0
Characteristics	IMAGE_FILE_32BIT_MACHINE IMAGE_FILE_EXECUTABLE_IMAGE
ImageVersion	0
SubsystemVersion	4
Subsystem	IMAGE_SUBSYSTEM_WINDOWS_GUI
File version	(None)
Product version	(None)
File flags	0000
Language code	Unknown
CharacterSet	Unknown
Company name	(None)
File description	(None)
Internal name	(None)
Legal Copyright	(None)
Production name	(None)
Production version	(None)

ssdeep Matches

No matches found.

PE Sections

Name	Virtual Address	Virtual Size	Raw Size	MD5	Entropy
.text	8192	94020	94208	3813db0d50813cb6bfd25f4ddb8bbbc5d	5.6
.reloc	106496	512	512	693bac427840a69136b15c59253a9438	4.839559

Packers/Compilers/Cryptos

(Heur)Protection: Anti analysis[Anti-dnSpy + Anti-ILSpy]

Imports (1)

- mscoree.dll
 - > _CorExeMain

TLP: WHITE

Relationships

7462acb6f1... imported 25da610be6acecf71bbe3a4e88c09f31ad07bdd252e
b30feef9debd9667c51

Description

There are various types of variants of njRAT, and attack tools deployed other than variants have been distributed in various versions such as Green edition, Dark edition, Danger edition, njRat-0.7d, njRAT-0.9g, etc. Various versions of njRAT tools are often used in conjunction with packaging technology because they have a variety of fancy technologies, bypass technologies, anti-debugging, and connection persistence technologies for each version.

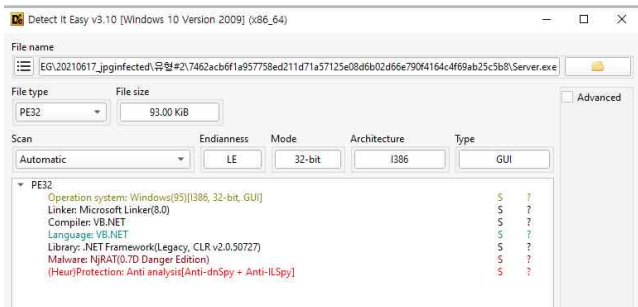


Figure 1 - check packer using Detect It Easy

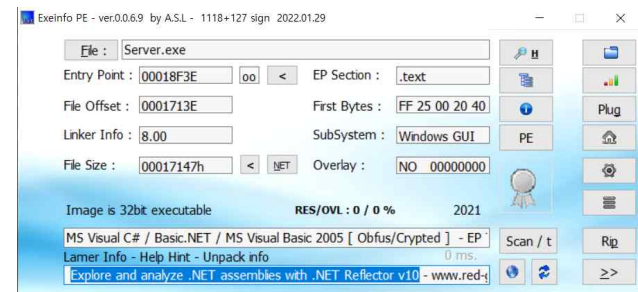


Figure 2 - check packer using Exeinfo PE

Detect It Easy (DIE) and Exeinfo PE tools were used to check whether malicious files to be analyzed were packaged, and anti-dnSpy and anti-ILSpy technologies provided by .Net framework were applied to the type of packer.

In the process of extracting the string value, a long word or string included in the file to be analyzed is displayed, providing a good clue to what you are trying to do or potential In-Current Objects (IOCs). You can also do this using the strings command, Only string data of the file to be analyzed were extracted using Ghidra. In addition, cross-verification was performed using binText and flare-floss tool.

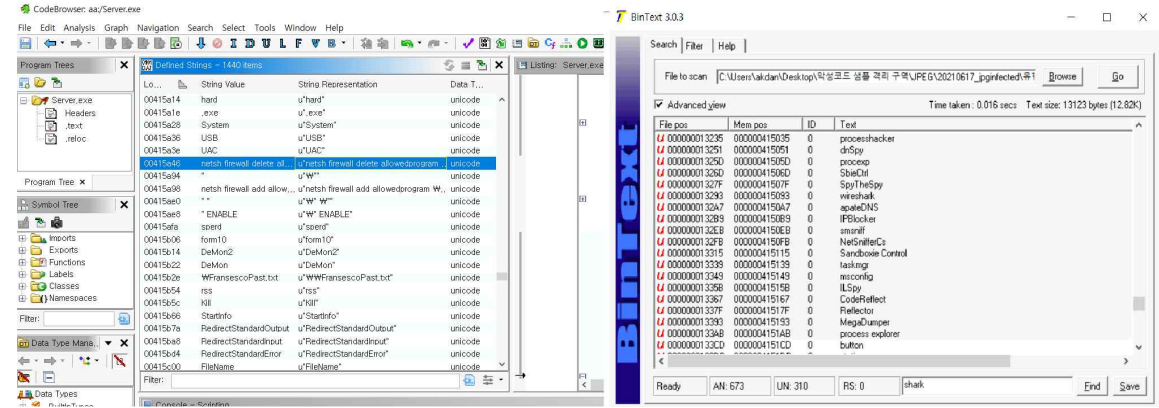


Figure 3 - check string values from malware file using Ghidra and BinText

Analyzing the extracted string values, I found some interesting strings from file.

Class	Value			
Encoded C2 Server address data	00415486	NTU1Mg==	u"NTU1Mg=="	unicode
	00415498	36f6c7682bfa79d619dca68...	u"36f6c7682bfa79d619dca68e72ce758d"	unicode
Malware version, Author's comment and several information				
Executable referenced				
Interesting commands	000000013C47	000000415A47	0	netsh firewall delete allowedprogram "
	000000013C99	000000415A99	0	netsh firewall add allowedprogram "
anti-debugging (disallow) program list	723	000000013235	000000415035	0 processhacker
	724	000000013251	000000415051	0 dnSpy
	725	00000001325D	00000041505D	0 procexp
	726	00000001326D	00000041506D	0 SbieCtrl
	727	00000001327F	00000041507F	0 SpyTheSpy
	728	000000013293	000000415093	0 Wireshark
	729	0000000132A7	0000004150A7	0 apateDNS
	730	0000000132B9	0000004150B9	0 IPBlocker
	731	0000000132EB	0000004150EB	0 smsniff
	732	0000000132FB	0000004150FB	0 NetSnifferCs
	733	000000013315	000000415115	0 Sandboxie Control
	734	000000013339	000000415139	0 taskmgr
	735	000000013349	000000415149	0 msconfig
	736	00000001335B	00000041515B	0 ILSpy
	737	000000013367	000000415167	0 CodeReflect
	738	00000001337F	00000041517F	0 Reflector
	739	000000013393	000000415193	0 MegaDumper
	740	0000000133AB	0000004151AB	0 process explorer

Among the extracted strings, the domain address of the encoded C2 server could be checked.

00415486	NTU1Mg==	u"NTU1Mg=="	unicode
00415498	36f6c7682bfa79d619dca68...	u"36f6c7682bfa79d619dca68e72ce758d"	unicode

Figure 4 - encoded C2 Server domain address data

The value of 'NTU1Mg==' located at the address 000000013687 is data encoded with pure Base 64. When this value is decoded, the value of '5552' is extracted. This may be guessed by the port number for connection with the C2 server, and as a result of cross-validation through TCPView, it was confirmed that the malicious file to be analyzed uses the same port to connect with the C2 server.

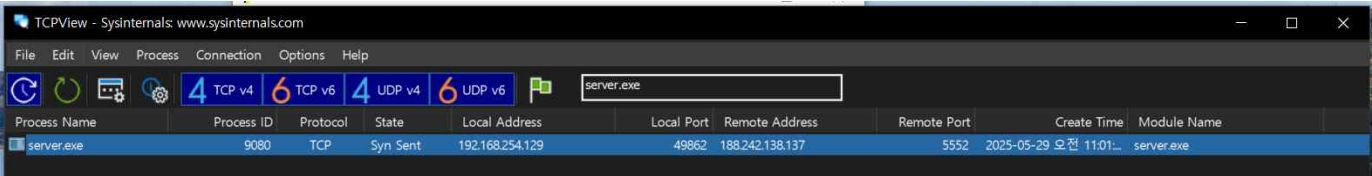


Figure 5 - Connection check results using TCPView for cross-validation of decoded port numbers

Original data (encoded)	decoded data
NTU1Mg==	5552
aGFraW0z	hakim3
i5kZG5zLm5ldA	ddns.net
Expected data result : <i>hakim3.ddns.net:5552</i>	

In the extracted value 'aGFraW0z*i5kZG5zLm5ldA!!', '!!' is determined as a substitution for bypassing the base64 padding (e.x.==). Therefore, the decoding is attempted based on the processed value 'aGFraW0z*i5kZG5zLm5ldA'.

The value of 'aGFraW0z' in the front part of the extracted value was decoded by Base64 and had the value of "hakim3", and the value of 'i5kZG5zM5ldA' in the rear part had the value of ".ddns.net ". Thus, the final value of the extracted data is "hakim3.ddns.net:5552 ".

188.242.138.137

Ports

- 5552 TCP

Connections

PID	Process	IP Domain	Domain	ASN	Country	Reputation
3400	server.exe	188.242.138.137:5552	hakim3.ddns.net	SkyNet Ltd.	RU	Malicious

Threats

PID	Process	Class	Message
3400	server.exe	Network Trojan was detected	ET TROJAN Generic njRAT/Bladabindi CnC Activity (II)

Whois

% This is the RIPE Database query service.
% The objects are in RPSL format.
%
% The RIPE Database is subject to Terms and Conditions.
% See <https://docs.db.ripe.net/terms-conditions.html>
% Note: this output has been filtered.
% To receive output for a database update, use the "-B" flag.
% Information related to '188.242.0.0 - 188.242.255.255'
% Abuse contact for '188.242.0.0 - 188.242.255.255' is 'abuse@sknt.ru'

inetnum: 188.242.0.0 - 188.242.255.255
netname: SkyNet
country: RU
admin-c: SKNT2-RIPE
tech-c: SKNT2-RIPE
status: ASSIGNED PA
mnt-by: MNT-SKNT
mnt-lower: MNT-SKNT
mnt-routes: MNT-SKNT
created: 2009-06-30T12:40:02Z
last-modified: 2022-08-31T17:53:57Z
source: RIPE

role: SKYNET NOC
address: SkyNet LLC

address: 192239 St. Petersburg
address: Russian Federation
phone: +7 (812) 386 20 20
remarks: -----
remarks: Routing and peering issues: tp2@hub.sknt.ru
remarks: Abuse and security: tp2@hub.sknt.ru
remarks: -----
abuse-mailbox: abuse@sknt.ru
admin-c: MK5687-RIPE
tech-c: MK5687-RIPE
nic-hdl: SKNT2-RIPE
mnt-by: MNT-SKNT
created: 2008-04-21T16:28:30Z
last-modified: 2020-05-16T12:43:27Z
source: RIPE # Filtered

% Information related to '188.242.128.0/18AS35807'

route: 188.242.128.0/18
descr: SkyNet Networks
origin: AS35807
mnt-by: MNT-SKNT
created: 2009-07-06T09:26:17Z
last-modified: 2009-07-06T09:26:17Z
source: RIPE

% This query was served by the RIPE Database Query Service version 1.117 (ABERDEEN)

Recommendations

Author recommends that users and administrators consider using the following best practices to strengthen the security posture of their organization's systems. Any configuration changes should be reviewed by system owners and administrators prior to implementation to avoid unwanted impacts. The following checklist reflects the author's personal best-practice recommendations for safeguarding endpoints and servers. Review and test any configuration change in a lab or staging environment before applying it to production systems.

- Maintain up-to-date antivirus signatures and engines.
- Keep operating system patches up-to-date.
- Disable File and Printer sharing services. If these services are required, use strong passwords or Active Directory authentication.
- Restrict users' ability (permissions) to install and run unwanted software applications. Do not add users to the local administrators group unless required.
- Enforce a strong password policy and implement regular password changes.
- Exercise caution when opening e-mail attachments even if the attachment is expected and the sender appears to be known.
- Enable a personal firewall on agency workstations, configured to deny unsolicited connection requests.
- Disable unnecessary services on agency workstations and servers.
- Scan for and remove suspicious e-mail attachments; ensure the scanned attachment is its "true file type" (i.e., the extension matches the file header).
- Monitor users' web browsing habits; restrict access to sites with unfavorable content.
- Exercise caution when using removable media (e.g., USB thumb drives, external drives, CDs, etc.).
- Scan all software downloaded from the Internet prior to executing.
- Maintain situational awareness of the latest threats and implement appropriate Access Control Lists (ACLs).

Additional information on malware incident prevention and handling can be found in National Institute of Standards and Technology(NIST) Special Publication 800-83, **"Guide to Malware Incident Prevention & Handling for Desktops and Laptops"**.

Contact Information

- Github (github.com/miho030)

Document FAQ

Q. Is it safe to deploy the IOCs from this report directly in a live environment?

A. Best practice is to apply the IOCs in a test or staging environment first, verify there are no false positives, and then roll them out to production in phases.

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A. You can legally download samples from public repositories and sandboxes such as VirusTotal, MalwareBazaar, and ANY.RUN. If you execute them locally, always use an isolated analysis environment (e.g., a virtual machine or Cuckoo sandbox).